

Terna Engineering College

Computer Engineering Department

Class: TE

Sem.: VI

Course: System Security Lab

PART A

(PART A : TO BE REFERRED BY STUDENTS)

Experiment No.7

A.1 Aim: Download and install nmap. Use it with different options to scan open ports, perform OSfingerprinting, do a ping scan, tcp port scan, udp port scan, xmas scan etc. A.2 Prerequisite:

1. Basic Knowledge of Ports, TCP, UDP, Ping

A.3 Outcome:

After successful completion of this experiment students will be able to

Install and use nmap and use it for gathering detailed network and remote host information. A.4 Theory:

Nmap (Network Mapper) is a security scanner originally written by Gordon Lyon (also known by his pseudonym Fyodor Vaskovich) used to discover hosts and services on a computer network, thus creating a "map" of the network. To accomplish its goal, Nmap sends specially crafted packets to the target host and then analyzes the responses. Unlike many simple port scanners that just send packets at some predefined constant rate, Nmap accounts for the network conditions (latency fluctuations, network congestion, the target interference with the scan) during the run. Also, owing to the large and active user community providing feedback and contributing to its features, Nmap has been able to extend its discovery capabilities beyond simply figuring out whether a host is up or down and which ports are open and closed; it can determine the operating system of the target, names and versions of the listening services, estimated uptime, type of device, and presence of a firewall.

Nmap features include:

- Host Discovery – Identifying hosts on a network. For example, listing the hosts which respond to pings or have a particular port open.
- Port Scanning – Enumerating the open ports on one or more target hosts.

- Version Detection – Interrogating listening network services listening on remote devices to determine the application name and version number.
- OS Detection –Remotely determining the operating system and some hardware characteristics of network devices.

Basic commands working in Nmap:

- For target specifications: `nmap<target's URL or IP with spaces between them>`
- For OS detection: `nmap -O <target-host's URL or IP>`
- For version detection: `nmap -sV<target-host's URL or IP>`

SYN scan is the default and most popular scan option for good reasons. It can be performed quickly, scanning thousands of ports per second on a fast network not hampered by restrictive firewalls. It is also relatively unobtrusive and stealthy since it never completes TCP connections.

A5.Procedure:

Installation of Nmap: \$

`sudo apt-get install nmap`

Commands:

- `nmap -sP<10.0.0.0/24>`

Ping scans the network, listing machines that respond to ping.

- FIN scan (`-sF`)

Sets just the TCP FIN bit.

- `-sV` (Version detection) .

Enables version detection, as discussed above. Alternatively, can use `-A`, which enables version detection among other things.

- `-sO` (IP protocol scan) .

IP protocol scan allows you to determine which IP protocols (TCP, ICMP, IGMP, etc.) are supported by target machines. This isn't technically a port scan, since it cycles through IP protocol numbers rather than TCP or UDP port numbers.

- `-O` (Enable OS detection) .

Enables OS detection, as discussed above. Alternatively, you can use `-A` to enable OS detection along with other things.

- `-p` port ranges (Only scan specified ports) .

This option specifies which ports you want to scan and overrides the default. Individual port numbers are OK, as are ranges separated by a hyphen (e.g. 1-1023). The beginning and/or end values of a range may be omitted, causing Nmap to use 1 and 65535, respectively.

- `--top-ports <integer of 1 or greater>`

Scans the N highest-ratio ports found in nmap-services file.

- `nmap -iflist` host interface and route information with nmap by using `--iflist` option.

PART B

(PART B : TO BE COMPLETED BY STUDENTS)

(Students must submit the soft copy as per following segments within two hours of the practical. The soft copy must be uploaded on the Blackboard or emailed to the concerned lab in charge faculties at the end of the practical in case there is no Black board access available)

Roll No. C54	Name: Quazi Md Moinuddin
Class : TE-C	Batch : C3
Date of Experiment: 3-3-25	Date of Submission :17-3-25
Grade :	

B.1 Output (add snapshot of output)

```
+ ~ nmap -sP 8.8.8.8
Starting Nmap 7.95 ( https://nmap.org ) at 2025-03-17 23:36 IST
Nmap scan report for dns.google (8.8.8.8)
Host is up (0.11s latency).
Nmap done: 1 IP address (1 host up) scanned in 0.16 seconds

+ ~ nmap -sF 8.8.8.8
You requested a scan type which requires root privileges.
QUITTING!

+ ~ nmap -sV 8.8.8.8
Nmap 7.95 ( https://nmap.org )
Usage: nmap [Scan Type(s)] [Options] {target specification}
TARGET SPECIFICATION:
  Can pass hostnames, IP addresses, networks, etc.
  Ex: scanme.nmap.org, microsoft.com/24, 192.168.0.1; 10.0.0-255.1-254
  -iL <inputfilename>: Input from list of hosts/networks
  -iR <num hosts>: Choose random targets
  --exclude <host1[,host2][,host3],...>: Exclude hosts/networks
  --excludefile <exclude_file>: Exclude list from file
HOST DISCOVERY:
  -sL: List Scan - simply list targets to scan
  -sn: Ping Scan - disable port scan
  -Pn: Treat all hosts as online -- skip host discovery
  -PS/PA/PY[portlist]: TCP SYN, TCP ACK, UDP or SCTP discovery to given ports
  -PE/PP/PM: ICMP echo, timestamp, and netmask request discovery probes
  -PO[protocol list]: IP Protocol Ping
  -n/-R: Never do DNS resolution/Always resolve [default: sometimes]
  --dns-servers <serv1[,serv2],...>: Specify custom DNS servers
  --system-dns: Use OS's DNS resolver
  --traceroute: Trace hop path to each host
SCAN TECHNIQUES:
  -sS/sT/sA/sW/sM: TCP SYN/Connect()/ACK/Window/Maimon scans
  -sU: UDP Scan
  -sN/sF/sX: TCP Null, FIN, and Xmas scans
  --scanflags <flags>: Customize TCP scan flags
  -sI <zombie host[:probeport]>: Idle scan
  -sV/sZ: SCTP INIT/COOKIE-ECHO scans
  -sO: IP protocol scan
  -b <FTP relay host>: FTP bounce scan
PORT SPECIFICATION AND SCAN ORDER:
  -p <port ranges>: Only scan specified ports
  Ex: -p22; -p1-65535; -p U:53,111,137,T:21-25,80,139,8080,S:9
  --exclude-ports <port ranges>: Exclude the specified ports from scanning
  -F: Fast mode - Scan fewer ports than the default scan
  -r: Scan ports sequentially - don't randomize
  --top-ports <number>: Scan <number> most common ports
  --port-ratio <ratio>: Scan ports more common than <ratio>
SERVICE/VERSION DETECTION:
```

```
--min-rate <number>: Send packets no slower than <number> per second
--max-rate <number>: Send packets no faster than <number> per second
FIREWALL/IDS EVASION AND SPOOFING:
-f; --mtu <val>: Fragment packets (optionally w/given MTU)
-D <decoy1,decoy2[,ME],...>: Cloak a scan with decoys
-S <IP_Address>: Spoof source address
-e <iface>: Use specified interface
-g/--source-port <portnum>: Use given port number
--proxies <url1[,url2],...>: Relay connections through HTTP/SOCKS4 proxies
--data <hex string>: Append a custom payload to sent packets
--data-string <string>: Append a custom ASCII string to sent packets
--data-length <num>: Append random data to sent packets
--ip-options <options>: Send packets with specified ip options
--ttl <val>: Set IP time-to-live field
--spooof-mac <mac address/prefix/vendor name>: Spoof your MAC address
--badsum: Send packets with a bogus TCP/UDP/SCTP checksum
OUTPUT:
-oN/-oX/-oS/-oG <file>: Output scan in normal, XML, sI<Ipt kIdDi3,
and Grepable format, respectively, to the given filename.
-oA <basename>: Output in the three major formats at once
-v: Increase verbosity level (use -vv or more for greater effect)
-d: Increase debugging level (use -dd or more for greater effect)
--reason: Display the reason a port is in a particular state
--open: Only show open (or possibly open) ports
--packet-trace: Show all packets sent and received
--iflist: Print host interfaces and routes (for debugging)
--append-output: Append to rather than clobber specified output files
--resume <filename>: Resume an aborted scan
--noninteractive: Disable runtime interactions via keyboard
--stylesheet <path/URL>: XSL stylesheet to transform XML output to HTML
--webxml: Reference stylesheet from Nmap.Org for more portable XML
--no-stylesheet: Prevent associating of XSL stylesheet w/XML output
MISC:
-6: Enable IPv6 scanning
-A: Enable OS detection, version detection, script scanning, and traceroute
--datadir <dirname>: Specify custom Nmap data file location
--send-eth/--send-ip: Send using raw ethernet frames or IP packets
--privileged: Assume that the user is fully privileged
--unprivileged: Assume the user lacks raw socket privileges
-V: Print version number
-h: Print this help summary page.
EXAMPLES:
nmap -v -A scanme.nmap.org
nmap -v -sn 192.168.0.0/16 10.0.0.0/8
nmap -v -iR 10000 -Pn -p 80
SEE THE MAN PAGE (https://nmap.org/book/man.html) FOR MORE OPTIONS AND EXAMPLES
An option is required for -s, most common are -sT (tcp scan), -sS (SYN scan), -sF (FIN scan), -sU (UDP scan) and -sn (Ping scan)
```

```
+ ~ █
```

```
MISC:
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```

```
+ ~ sudo nmap -sO 8.8.8.8
Password:
Starting Nmap 7.95 ( https://nmap.org ) at 2025-03-17 23:37 IST
Nmap scan report for dns.google (8.8.8.8)
Host is up (0.12s latency).
Not shown: 254 open/filtered n/a protocols (no-response)
PROTOCOL STATE SERVICE
1 open icmp
6 open tcp

Nmap done: 1 IP address (1 host up) scanned in 7.00 seconds
```

```
+ ~ nmap -O 8.8.8.8
nmap: unrecognized option '-O'
See the output of nmap -h for a summary of options.
```

```
+ ~ nmap -O 8.8.8.8
TCP/IP fingerprinting (for OS scan) requires root privileges.
QUITTING!
```

```
+ ~ ) nmap-p 80 8.8.8.8
zsh: parse error near `)'
```

```
+ ~ nmap-p 80 8.8.8.8
zsh: command not found: nmap-p
```

```
+ ~ nmap -p 80 8.8.8.8
nmap: unrecognized option '-p'
See the output of nmap -h for a summary of options.
```

```
+ ~ nmap 80 8.8.8.8
Starting Nmap 7.95 ( https://nmap.org ) at 2025-03-17 23:38 IST
█
```

```
SEE THE MAN PAGE (https://nmap.org/book/man.html) FOR MORE OPTIONS AND EXAMPLES
An option is required for -s, most common are -sT (tcp scan), -sS (SYN scan), -sF (FIN scan), -sU (UDP scan) and -sn (Ping scan)

+ ~ sudo nmap -s0 8.8.8.8
Password:
Starting Nmap 7.95 ( https://nmap.org ) at 2025-03-17 23:37 IST
Nmap scan report for dns.google (8.8.8.8)
Host is up (0.12s latency).
Not shown: 254 open/filtered n/a protocols (no-response)
PROTOCOL STATE SERVICE
1      open  icmp
6      open  tcp

Nmap done: 1 IP address (1 host up) scanned in 7.00 seconds

+ ~ nmap -0 8.8.8.8
nmap: unrecognized option '-0'
See the output of nmap -h for a summary of options.

+ ~ nmap -O 8.8.8.8
TCP/IP fingerprinting (for OS scan) requires root privileges.
QUITTING!

+ ~ ) nmap-p 80 8.8.8.8
zsh: parse error near `)'

+ ~ nmap-p 80 8.8.8.8
zsh: command not found: nmap-p

+ ~ nmap -p 80 8.8.8.8
nmap: unrecognized option '-p'
See the output of nmap -h for a summary of options.

+ ~ nmap 80 8.8.8.8
Starting Nmap 7.95 ( https://nmap.org ) at 2025-03-17 23:38 IST
Nmap scan report for dns.google (8.8.8.8)
Host is up (0.046s latency).
Not shown: 994 filtered tcp ports (no-response)
PORT      STATE SERVICE
21/tcp    open  ftp
53/tcp    open  domain
113/tcp   closed ident
443/tcp   open  https
554/tcp   open  rtsp
1723/tcp  open  pptp

Nmap done: 2 IP addresses (1 host up) scanned in 47.77 seconds

+ ~ █

+ ~ nmap -iflist
Starting Nmap 7.95 ( https://nmap.org ) at 2025-03-17 23:39 IST
*****INTERFACES*****
DEV      (SHORT)  IP/MASK      TYPE      UP      MTU      MAC
lo0      (lo0)     127.0.0.1/8  loopback  up      16384
lo0      (lo0)     ::1/128      loopback  up      16384
lo0      (lo0)     fe80::1/64   loopback  up      16384
gif0     (gif0)    (none)/0     point2point down    1280
stf0     (stf0)    (none)/0     other      down    1280
anp11    (anp11)   (none)/0     ethernet  up      1500    66:09:D8:EB:86:B8
anp10    (anp10)   (none)/0     ethernet  up      1500    66:09:D8:EB:86:B7
en3      (en3)     (none)/0     ethernet  up      1500    66:09:D8:EB:86:97
en4      (en4)     (none)/0     ethernet  up      1500    66:09:D8:EB:86:98
en1      (en1)     (none)/0     ethernet  up      1500    36:DE:B5:29:2F:40
en2      (en2)     (none)/0     ethernet  up      1500    36:DE:B5:29:2F:44
ap1      (ap1)     (none)/0     ethernet  up      1500    66:0A:85:71:E8:C8
en0      (en0)     192.168.96.230/24  ethernet  up      1500    B6:C1:40:51:9A:37
en0      (en0)     fe80::452:a13c:af11:3c39/64  ethernet  up      1500    B6:C1:40:51:9A:37
en0      (en0)     2409:40c0:106a:956e:c5d:9ca8:e032:3f1d/64  ethernet  up      1500    B6:C1:40:51:9A:37
en0      (en0)     2409:40c0:106a:956e:8f:8e6f:a045:8ca6/64  ethernet  up      1500    B6:C1:40:51:9A:37
bridge0  (bridge0) (none)/0     ethernet  up      1500    36:DE:B5:29:2F:40
awdl0    (awdl0)   (none)/0     ethernet  up      1500    36:50:B7:B0:E5:CF
awdl0    (awdl0)   fe80::3450:b7ff:feb0:e5cf/64  ethernet  up      1500    36:50:B7:B0:E5:CF
llw0     (llw0)    (none)/0     ethernet  up      1500    36:50:B7:B0:E5:CF
llw0     (llw0)    fe80::3450:b7ff:feb0:e5cf/64  ethernet  up      1500    36:50:B7:B0:E5:CF
utun0    (utun0)   (none)/0     point2point up      1380
utun0    (utun0)   fe80::782a:64da:4f42:7de4/64  point2point up      1380
utun1    (utun1)   (none)/0     point2point up      2000
utun1    (utun1)   fe80::59c4:a3cf:5a96:3b5c/64  point2point up      2000
utun2    (utun2)   (none)/0     point2point up      1000
utun2    (utun2)   fe80::ce81:b1c:bd2c:69e/64    point2point up      1000
utun3    (utun3)   (none)/0     point2point up      1500
utun3    (utun3)   fe80::6627:1e84:fed2:7f72/64  point2point up      1500

*****ROUTES*****
DST/MASK      DEV      METRIC GATEWAY
255.255.255.255/32  en0      0       192.168.96.45
35.212.73.115/32   en0      0       192.168.96.45
3.160.188.128/32   en0      0       192.168.96.45
3.229.220.154/32   en0      0       192.168.96.45
4.255.57.234/32    en0      0       192.168.96.45
8.8.8.8/32         en0      0       192.168.96.45
13.89.179.11/32    en0      0       192.168.96.45
13.201.189.139/32  en0      0       192.168.96.45
17.8.135.181/32    en0      0       192.168.96.45
17.23.112.11/32    en0      0       192.168.96.45
17.156.129.37/32   en0      0       192.168.96.45
17.242.13.4/32     en0      0       192.168.96.45
```

```
2603:1046:1406::5/128      en0  0      fe80:b::785f:1cff:fe10:5ac
2606:4700:3030::6815:2001/128 en0  0      fe80:b::785f:1cff:fe10:5ac
2606:4700:3030::6815:3001/128 en0  0      fe80:b::785f:1cff:fe10:5ac
2606:4700:3030::6815:4001/128 en0  0      fe80:b::785f:1cff:fe10:5ac
2606:4700:3030::6815:5001/128 en0  0      fe80:b::785f:1cff:fe10:5ac
2a01:b740:a14:1000::152/128  en0  0      fe80:b::785f:1cff:fe10:5ac
2a01:b740:a14:2000::132/128  en0  0      fe80:b::785f:1cff:fe10:5ac
2a01:b740:a14:f000::197/128  en0  0      fe80:b::785f:1cff:fe10:5ac
2a03:2880:f22f:c6:face:b00c:0:7260/128 en0  0      fe80:b::785f:1cff:fe10:5ac
2a03:2880:f358:120:face:b00c:0:167/128 en0  0      fe80:b::785f:1cff:fe10:5ac
2a03:2880:f358:121:face:b00c:0:7260/128 en0  0      fe80:b::785f:1cff:fe10:5ac
fe80:1::/64                  lo0   0      fe80:1:::1
fe80:11::/64                 utun2  0      fe80:11::ce81:b1c:bd2c:69e
2409:40c0:106a:956e::/64    en0   0
fe80:b::/64                  en0   0
fe80:d::/64                  awdl0  0
fe80:f::/64                  utun0  0      fe80:f::782a:64da:4f42:7de4
fe80:12::/64                 utun3  0      fe80:12::6627:1e84:fed2:7f72
fe80:10::/64                 utun1  0      fe80:10::59c4:a3cf:5a96:3b5c
ff02:12::/32                 utun3  0      fe80:12::6627:1e84:fed2:7f72
ff02:b::/32                  en0   0
ff02:d::/32                  awdl0  0
ff02:11::/32                 utun2  0      fe80:11::ce81:b1c:bd2c:69e
ff02:10::/32                 utun1  0      fe80:10::59c4:a3cf:5a96:3b5c
ff02:f::/32                  utun0  0      fe80:f::782a:64da:4f42:7de4
ff01:1::/32                  lo0   0      ::1
ff01:b::/32                  en0   0
ff01:11::/32                 utun2  0      fe80:11::ce81:b1c:bd2c:69e
ff01:12::/32                 utun3  0      fe80:12::6627:1e84:fed2:7f72
ff01:10::/32                 utun1  0      fe80:10::59c4:a3cf:5a96:3b5c
ff01:f::/32                  utun0  0      fe80:f::782a:64da:4f42:7de4
ff02:1::/32                  lo0   0      ::1
ff01:d::/32                  awdl0  0
ff00::/8                     en0   0
ff00::/8                     utun1  0      fe80:10::59c4:a3cf:5a96:3b5c
ff00::/8                     utun2  0      fe80:11::ce81:b1c:bd2c:69e
ff00::/8                     awdl0  0
ff00::/8                     utun3  0      fe80:12::6627:1e84:fed2:7f72
ff00::/8                     llw0   0
ff00::/8                     lo0   0      ::1
ff00::/8                     utun0  0      fe80:f::782a:64da:4f42:7de4
::/0                          utun1  0      fe80:10::
::/0                          en0   0      fe80:b::785f:1cff:fe10:5ac
::/0                          utun2  0      fe80:11::
::/0                          utun0  0      fe80:f::
::/0                          utun3  0      fe80:12::

+ ~ █

Last login: Mon Mar 17 23:31:48 on ttys000
/Users/mdmoinuddinquazi/.zshenv.:1: no such file or directory: /Users/mdmoinuddinquazi/.cargo/env

+ ~ █

+ ~ nmap --top-ports 1000 8.8.8.8
Starting Nmap 7.95 ( https://nmap.org ) at 2025-03-17 23:39 IST
Nmap scan report for dns.google (8.8.8.8)
Host is up (0.057s latency).
Not shown: 995 filtered tcp ports (no-response)
PORT      STATE SERVICE
21/tcp    open  ftp
53/tcp    open  domain
443/tcp   open  https
554/tcp   open  rtsp
1723/tcp  open  pptp

Nmap done: 1 IP address (1 host up) scanned in 62.62 seconds

+ ~ █
```

B.2. Commands / tools used with syntax:

- 1) nmap -sP 8.8.8.8
- 2) nmap -sF 8.8.8.8

- 3) `nmap -s V 8.8.8.8`
- 4) `sudo nmap -sO 8.8.8.8`
- 5) `nmap -0 8.8.8.8`
- 6) `nmap-p 80 8.8.8.8`
- 7) `nmap --top-ports 1000 8.8.8.8`
- 8) `nmap -iflist`

B.3 Question of Curiosity:

1. What are the ethical considerations when using Nmap for scanning networks?
2. What precautions should be taken to avoid detection when using Nmap for network scanning?
3. What is the purpose of OS fingerprinting in network scanning?
4. How does an Xmas scan work, and what does it detect?
5. What is the difference between TCP and UDP port scanning?
6. How can Nmap be used to detect firewalls and filtering devices?

C-54

classmate

Date _____
Page _____

- To filter SSH, you would look traffic on port 22, but the content will be encrypted.
tcp.port = 22

Exp-7 C54

Q1) What are the ethical considerations when using Nmap for scanning networks.

-
1. Permissions - Always obtain explicit permission before scanning networks that are not your own.
 2. Impact on Target Network - Scanning, especially aggressive or full-scale scanning can create an load on the network, it can create alerts, slow down services.
 3. Privacy concerns - Network scanning can expose sensitive information.
 4. Compliance with laws & regulations - In many regions, there are strict laws about accessing or probing systems without permission.

2) What precautions should be taken to avoid detection when using Nmap for network scanning?

→ When performing a network scan with Nmap, especially in environments where you're to avoid detection, there are several strategies you can use:

1. Use stealth scans (Avoid Detection),

° TCP SYN Scan: Use the SYN scan (-SS) also known as the "half-open scan". It can only send a SYN packet & wait for a response without completing the TCP handshake. This makes it more difficult to detect because the connⁿ is never fully established.
\$ nmap -SS [target]

- Sleeper Scan: The sleeper scan (-SI) is a stealth scan that uses a third-party system to send packets to the target. This makes it harder to associate the scan with your IP address, making it more difficult to detect.

eg. `nmmap -SI [source ip] [target]`

2. Slow Down the Scan:

• Use timing options like `-T0` & `-T3` to slow down your scan & reduce the likelihood of triggering IDS/IPS alerts.

eg `nmmap -T1 -SS [target]`

3. Use Fragmentation

• Packet fragmentation (`-f`) can be used to break up packets into smaller pieces. This makes it harder for network devices (eg. firewalls) to detect the full packet structure.

eg `nmmap -f [target]`.

3) What is the ~~same~~ purpose of OS fingerprinting in network scanning?

→ OS fingerprinting is a technique used in network scanning to identify the OS running on the target device or host. This is an important step in reconnaissance & vulnerability assessment.

The purpose of OS fingerprinting is:

1. Identify the Target System: By determining the OS, attackers or security analysts can gain a better understanding of the system's architecture & vulnerabilities that may be specific to that OS.
2. Tailor Exploits: Once the OS is identified, an attacker can tailor their attack strategy based on known vulnerabilities specific to the operating system, or a security analyst

can assess the potential risks based on OS type.

3. ~~Assess~~ Security Posture: OS fingerprinting can help administrators evaluate the security posture of the network of what OS version & configurations are in use, which could potentially expose them to certain vulnerabilities.

4. Network mapping: Identifying the OS of different devices on a network helps in building a network topology & understanding the device characteristics.

B.4 Conclusion:

(Write appropriate conclusion.)

We've effectively showed how to use Nmap for network reconnaissance, including port scanning and OS fingerprinting and completed the experiment. By understanding the principles of port scanning and using Nmap and other relevant tools, we were able to gain information about vulnerability discovery and network security assessment.